Do Students Agree with LLM Feedback on Their Own Code? A Within-Subject Study of Four Feedback Configurations in Software Project Courses

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Abstract. Oral defense (viva) in software engineering capstone projects is essential for assessing students' understanding of business logic and codebase architecture. However, manual assessment is labor-intensive and may lead to inconsistent evaluations. In contrast, off-the-shelf large language models (LLMs) may exhibit code-context hallucinations, remain vulnerable to prompt injection, and produce sycophantic evaluations. This paper presents the pedagogical evaluation engine of **PACE**, an AI-assisted viva assessment system that combines complexity-aware hybrid ground-truth extraction—which selects between raw source code and Mermaid-based control-flow and data-flow graphs according to a node-count threshold ($\tau = 10$)—with a three-layer guardrail architecture and XML-based delimiter sandboxing. To mitigate single-model sycophancy, we propose a role-decoupled **Multi-LLM Ensemble** in *Researching Mode*, which uses **DeepSeek-V3** as an objective *Code Inspector* and **Groq (`openai/gpt-oss-120b`)** as a *Pedagogical Synthesizer and Rubric Grader*. We evaluate this approach alongside three single-LLM formative feedback paradigms (*Sandwich*, *Rubric*, and *Explainable*) and, for *Researching*, an intended two-stage model pipeline. Across 344 rated instances, 79.9% of ratings endorsed the AI's judgement, 68.0% endorsed future use and 41.9% endorsed motivation. All six 95% repository-bootstrap contrasts on the direct Acceptance item included zero, without establishing equivalence. Exploratory Sandwich–Explainable actionability differed by 0.37 points (approximate simultaneous 95% sensitivity band [0.02, 0.72] across 90 item contrasts); this bundle comparison does not isolate next steps. A secondary multidomain composite averaged 4.07/5 (95% CI [3.90, 4.23]); its small mean differences coexisted with distinct item profiles. The observational raw-code versus graph comparison on Acceptance was inconclusive ($b = 0.03$, 95% CI $[-0.18, 0.26]$). These findings concern reported perceptions, not expert-verified correctness, observed revision or learning gains.

Keywords: Automated Viva Assessment · Multi-LLM Ensemble · Formative Feedback · Pedagogical AI Guardrails · Software Engineering Education.

1 Introduction

Formative feedback is widely recognized as an important factor in improving student learning. However, its effectiveness depends largely on how learners understand, interpret, and act on the guidance they receive [11,31]. Technically accurate feedback does not necessarily lead to better understanding or meaningful revisions [1,34]. Research on feedback literacy emphasizes that students need to interpret feedback, make judgments about their own work, and use the feedback to guide their actions. These activities are essential parts of the feedback process, rather than simply outcomes that occur after feedback is provided [1,34]. In human-computer interaction, users' acceptance of automated feedback and willingness to act on it are also influenced by its perceived usefulness [4,33], as well as by psychological factors such as trust and users' tendency to accept or reject algorithmic judgments [22,6,26].

With the rapid development of generative artificial intelligence, large language models (LLMs) are increasingly used in computing education to identify code issues and provide formative feedback at scale [16,5,3]. However, general-purpose LLMs may generate inaccurate feedback when they lack sufficient context about the actual student codebase, including cases in which they misinterpret code or produce unsupported explanations [13,19]. In addition, deploying LLMs in educational assessment requires appropriate safeguards to maintain pedagogical boundaries and reduce the risk of misuse, including prompt injection attacks [25]. Recent studies in programming education and higher education have also found that students' perceptions of AI-generated feedback vary across several dimensions, including technical accuracy, level of detail, the availability of supporting evidence, and trust in comparison with feedback from human experts [9,35,27,8].

Two research gaps motivate this study:

- **Gap 1 (Automated Viva Assessment and Hybrid Ground-Truth Contexts in Team Capstone Projects):** Software engineering capstone projects often involve shared codebases developed collaboratively by student teams. Traditional input-output autograders can assess whether a program produces the expected results, but they are less suited to determining whether individual students understand the code and its architecture. Although asking students questions about their own programs has been explored as a way to assess code understanding [24], the use of automated, scenario-based oral defenses (viva) that combine multi-language static analysis with complexity-aware routing between different code representations remains underexplored.
- **Gap 2 (Formative Feedback Formats and Single-Model Sycophancy):** Using a single LLM as an automated evaluator (*LLM-as-a-Judge*) can introduce biases, including a tendency to favor certain responses or follow patterns that affect grading consistency [36]. In practical systems, AI-generated feedback may combine praise, rubric-based assessment, code evidence, and model orchestration in different configurations. These configurations reflect different approaches to formative feedback: the *Sandwich* format balances positive and corrective comments [28]; *Rubric* uses explicit scoring criteria to support consistent evaluation [15]; *Explainable* emphasizes evidence and transparency [18]; and the *Multi-LLM Ensemble* uses multiple

models in coordinated roles to improve evaluation quality and reduce dependence on a single model's judgment [7,36]. However, comparative within-subject studies of these feedback formats, based on students' actual capstone project codebases and focused on learners' perceptions, remain limited.

To address these gaps, we present **PACE** (*Project Assistance and Course Evaluation*), an AI-assisted viva assessment system deployed at FPT University. PACE uses multi-language static analysis tools, including Roslyn for C#/.NET and Tree-sitter for Java, JavaScript, and TypeScript, to extract control-flow and data-flow information and represent it as graphs. A complexity-aware context router selects raw source code (`Raw`) when the number of distinct function nodes, N_w, is at most 10 ($N_w \leq 10$), and Mermaid flow graphs (`Graph`) when $N_w > 10$. These representations are used to generate scenario-based questions grounded in the project code through OpenAI `gpt-6-luna`. To support safe evaluation and feedback generation, PACE incorporates a three-layer safety guardrail architecture comprising input, execution, and output guardrails, together with XML-based delimiter sandboxing. The delimiters distinguish the ground-truth context, exam question, suggested answer, and student answer using the tags `<ground_truth_context>`, `<exam_question>`, `<suggested_answer>`, and `<student_answer>`. These safeguards are designed to reduce the risk of prompt injection and keep the evaluation process aligned with the intended task. PACE supports four formative feedback formats: three single-LLM configurations powered by DeepSeek-V3 (*Sandwich*, *Rubric*, and *Explainable*) and a role-decoupled *Researching* mode, referred to as the *Multi-LLM Ensemble*. In this mode, DeepSeek-V3 acts as the *Code Inspector*, while Groq (`openai/gpt-oss-120b`) serves as the *Pedagogical Synthesizer and Rubric Grader*.

We conducted a randomized, student-blinded, within-subject experiment in which students answered scenario-based questions about business rules in different modules of their team's project repositories. The feedback formats were randomly assigned without replacement, and students rated each feedback instance using a 15-item, five-point Likert-scale instrument in which all items were scored in the same direction. The instrument was developed with reference to the Feedback Perceptions Questionnaire (FPQ), research on explainable AI (XAI) and argument persuasiveness, psychological safety, and the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT). We also considered established methods for assessing measurement reliability and identifying potential response and method biases [2,12,29,20,32]. This study addresses three research questions:

RQ1 How do students evaluate the content quality, perceived usefulness, and overall acceptance of LLM-generated feedback on viva responses about their own capstone project codebases?

RQ2 How do students' agreement rates and multidomain ratings differ across the four feedback formats (*Sandwich*, *Rubric*, *Explainable*, and *Researching/Multi-LLM Ensemble*) based on non-parametric tests and pairwise comparisons with corrections for multiple testing [10,14,21]?

RQ3 To what extent are student acceptance and perceived feedback quality associated with the use of raw source-code contexts ($N_w \leq 10$) versus Mermaid flow-graph contexts ($N_w > 10$) in the deployed system?

The main contribution of this work is an integrated AI-assisted viva assessment system with multiple safety safeguards, together with an empirical evaluation of different formative feedback formats for capstone project defense preparation. This study extends existing approaches to automated program questioning [24] to team-based project defense and examines students' agreement with AI judgments separately from their perceptions of feedback actionability, learning-related motivation, and intention to use the system. The findings represent learners' self-reported perceptions under a controlled experimental design and statistical analysis. They do not directly establish expert-verified technical accuracy, actual revision behavior, or long-term improvements in learning outcomes.

The remainder of this paper is structured as follows: Section 2 reviews related literature; Section 3 presents the methodology and architecture; Section 4 details the experimental evaluation; Section 5 discusses findings; Section 6.1 examines threats to validity; and Section 7 concludes the paper.

2 Related Work

This work draws on three main research areas: automated formative feedback in computing education, student acceptance and perceptions of AI-generated guidance, and pedagogical feedback configurations in software engineering assessment.

2.1 LLM-Generated Feedback in Programming Education

Automated assessment in computing education predates the emergence of generative AI, with early diagnostic tools focusing primarily on static error detection rather than actionable next steps [17]. While recent evaluations demonstrate that off-the-shelf LLMs can effectively identify issues in code, they remain susceptible to hallucinating nonexistent defects or suggesting misleading fixes [13,19,3]. To address these safety and reliability concerns, specialized systems such as *CodeHelp* have demonstrated the potential of incorporating guardrails to provide scalable programming assistance without directly supplying complete solutions [25]. Building on these foundations, our approach shifts the evaluation focus from isolated, beginner-level coding snippets to comprehensive explanations of business logic and system architecture across evolving, multi-language team repositories.

2.2 Student Perceptions and Acceptance of AI Feedback

Understanding how learners respond to automated feedback requires examining established technology acceptance and trust frameworks. According to the Technology Acceptance Model (TAM) and UTAUT, perceived usefulness and ease of use are important predictors of technology acceptance and use [4,33], whereas appropriate reliance on automation requires calibrating trust to actual system reliability [22]. Psychological phenomena such as algorithm aversion, in which users prematurely abandon algorithmic recommendations after observing errors, coexist with instances of algorithm appreciation [6,26]. In computing education, empirical studies have revealed diverse learner preferences and perceptions: Zhang et al. documented student preferences for feedback

that explicitly references source code [35], Er et al. identified continued advantages of human instructor feedback [8], and Nazaretsky et al. demonstrated that student trust is sensitive to the perceived source of AI-generated feedback [27]. These findings highlight the importance of evaluating multiple dimensions of learner perception, including acceptance, behavioral intention, trust, and affective responses, alongside technical correctness.

2.3 Pedagogical Feedback Configurations and Multi-Model Orchestration

Different formats of formative feedback are grounded in distinct pedagogical principles. Sandwich feedback (*Sandwich*), which places constructive criticism between supportive remarks, has been shown in controlled studies to improve learners' affective responses, even when performance gains remain modest [28]. Explicit scoring rubrics (*Rubric*) establish transparent evaluation criteria and promote grading consistency [15], while evidence-based explanations (*Explainable*) can support appropriate trust calibration by providing transparent references to source code [18]. To mitigate grading biases and sycophancy observed in LLM-based evaluations (*LLM-as-a-Judge*) [36], multi-agent debate and multi-model coordination architectures have been proposed to improve factual accuracy and objectivity in reasoning [7,36]. However, existing literature offers limited evidence from rigorous, randomized within-subject evaluations comparing these distinct formative feedback approaches, particularly when implemented through role-decoupled multi-model ensembles operating on authentic student capstone codebases.

Position Relative to Prior Literature. While prior work has used automated comprehension checks to assess program understanding [24], our work extends this approach to a structured viva defense preparation workflow involving collaborative team repositories. Unlike studies focusing on isolated CS1 programming tasks or hypothetical model-generated code snippets [35,23], PACE investigates student perceptions of multiple AI feedback formats in the context of authentic, multi-language capstone projects. Furthermore, by implementing a complexity-aware hybrid context router ($\tau = 10$), a three-layer guardrail architecture with XML delimiter sandboxing, and a role-decoupled Multi-LLM Ensemble, this study provides an empirical evaluation of guardrailed, multi-format formative feedback in software engineering education.

3 The PACE System

Figure 1 summarizes the PACE pipeline, including repository analysis, complexity-aware context routing, question generation, and multi-format pedagogical evaluation.

3.1 Multi-Language Static Analysis and Complexity-Aware Context Routing

For each submitted team repository, PACE clones the target branch and parses C#/.NET source code using Roslyn and Java (Spring Boot), JavaScript, and TypeScript (Express, NestJS, Hono) source code using tree-sitter. Extracted API endpoints and controllers are grouped into user-facing *business functions* (e.g., "budget management" or "asset

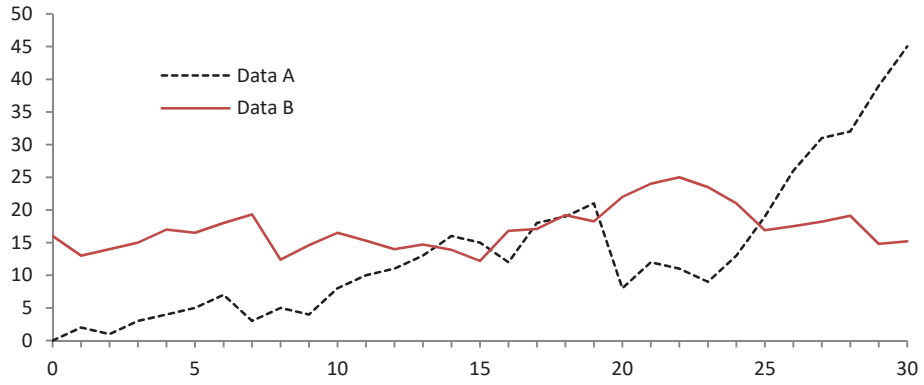


Fig. 1. Overview of the PACE system pipeline, from repository analysis to multi-format feedback.

auction") through metadata analysis performed by Gemini 3.1 Flash Lite, subject to instructor review and confirmation.

To assess code complexity and select an appropriate context representation, PACE computes the number of distinct function nodes N_w in each business function's call graph, derived from entry points and directed call relationships. A complexity-aware hybrid router then selects the context representation. When $N_w \leq 10$, the system retrieves raw source code (`Raw`) to preserve fine-grained details of conditional expressions. When $N_w > 10$, it uses a Mermaid control- and data-flow graph (`Graph`) containing ordered edges, branch conditions, and return types to reduce token overhead and syntactic noise. Scenario-based questions are generated for each business function using OpenAI `gpt-6-luna` (temperature 0.3, strict JSON schema output). Question generation follows a business-language-only constraint that prohibits technical keywords (such as *Controller*, *DTO*, or *SQL*). Questions are organized into three difficulty levels based on exception branches: Happy Path, Single Exception, and Double Exception.

3.2 Guardrailed Multi-Format Evaluation Engine and Multi-LLM Ensemble

When a student submits $k = 4$ scenario answers across different business functions, PACE uses an atomic database transaction (< 200 ms) with row-level locking (`SELECT FOR UPDATE`) to maintain data integrity before invoking external LLMs. Each submission is processed through a three-layer AI guardrail architecture: (1) an *Input Guardrail* that applies 17 blacklist-based checks for prompt injection and jailbreak attempts in English and Vietnamese; (2) an *Execution Guardrail* that blocks malicious scripts and binary payloads using compiled regular expressions and sanitizes Unicode control characters; and (3) an *Output Guardrail* that validates strict JSON structure and checks for potential API key leakage.

To reduce the risk of indirect prompt injection and reinforce the instruction hierarchy, the evaluation engine separates input components using XML delimiters:

```
<ground_truth_context> {Raw code or Mermaid graph} </ground_truth_context>
```

```
<exam_question> {Sanitized scenario question} </exam_question>
<suggested_answer> {Reference suggested answer} </suggested_answer>
<student_answer> {Sanitized student answer} </student_answer>
```

The system supports four feedback formats, assigned through a random permutation without replacement within each student ($\mathcal{F}$ = {Sandwich, Rubric, Explainable, Researching}), ensuring that every student encounters each configuration exactly once:

- **Sandwich**: Feedback structured into positive reinforcement, constructive criticism, and actionable next steps, generated by DeepSeek-V3 (`deepseek-chat`, temperature 0.2).
- **Rubric**: Quantitative feedback assessing technical competencies using a five-point scoring scale and detailed commentary, generated by DeepSeek-V3.
- **Explainable**: Evidence-based feedback incorporating explicit source-code citations (`code_evidence`) and explanations of logical reasoning (`logic_explanation`), generated by DeepSeek-V3.
- **Researching (Multi-LLM Ensemble)**: A role-decoupled, two-stage architecture designed to mitigate grading biases and sycophancy associated with single-model evaluation. In Stage 1, DeepSeek-V3 serves as a *Code Inspector*, extracting code evidence and identifying logical inconsistencies. In Stage 2, Groq (`openai/gpt-oss-120b`) serves as a *Pedagogical Synthesizer and Rubric Grader*, integrating the inspection report from DeepSeek-V3 to generate comprehensive pedagogical feedback and rubric scores. A multistage fault-tolerance mechanism handles rate-limit errors (HTTP 429) through exponential backoff and falls back to single-model processing if downstream validation fails.

3.3 Empirical Data Collection and Survey Instrumentation

After receiving feedback, students evaluate each feedback instance through a web interface using a 15-item questionnaire with a five-point Likert response scale (1 = Strongly disagree to 5 = Strongly agree). All items are positively keyed. Drawing on the Feedback Perceptions Questionnaire (FPQ), explainable AI trustworthiness, argument persuasiveness, psychological safety, and TAM/UTAUT frameworks, the 15 items assess information and pedagogical quality (`ACCURACY`, `SPECIFICITY`, `CLARITY`), explainability (`TRANSPARENCY`), usefulness and actionability (`USEFULNESS`, `ACTIONABILITY`), trust and fairness (`COMPETENCE TRUST`, `RELIABILITY`, `FAIRNESS`), affect and psychological safety (`AFFECT`, `PSYCHOLOGICAL SAFETY`), and behavioral acceptance (`ACCEPTANCE`, `PERSUASIVENESS`, `WILLINGNESS`, `BEHAVIORAL INTENTION`). The data export pipeline automatically excludes load-testing accounts (identified by MSSV values beginning with `K6`) to ensure that the resulting datasets contain only eligible student responses.

4 Experimental Design

4.1 Context and Participants

The empirical evaluation was conducted in October 2026 in undergraduate software engineering capstone courses at FPT University, Vietnam. The research team invited final-year software engineering students to use PACE to prepare for their oral defenses (vivas),

allowing them to practice explaining their teams' collaborative codebases. Participation was entirely voluntary, and students were explicitly assured that declining to participate would not affect their academic standing.

The resulting dataset comprises responses from 86 undergraduate students working across 28 distinct project repositories. Collectively, they submitted 344 scenario-based viva answers ($k = 4$ responses per student, each addressing a different business function). The answers were written in Vietnamese, with an average length of approximately 51 whitespace-delimited words per response, and were collected between October 20 and October 26, 2026.

4.2 Survey Instrumentation

After submitting each answer and receiving AI-generated feedback, students evaluated the corresponding feedback instance through a web interface using a 15-item questionnaire with a five-point Likert response scale (1 = Strongly disagree to 5 = Strongly agree). All fifteen items were positively keyed, with higher scores indicating more favorable perceptions. Drawing on established principles of formative feedback [31], technology acceptance frameworks (TAM/UTAUT) [4,33], trust in automation [22], argument persuasiveness, and psychological safety, the instrument assesses six conceptual constructs, as detailed in Table 1.

4.3 Dataset and Experimental Procedure

The final dataset contains 344 complete feedback-rating records from 86 students, each of whom submitted one set of four answers. Every student evaluated all four feedback formats (*Sandwich*, *Rubric*, *Explainable*, and *Researching*), which were randomly assigned to their four answers without replacement. This design yielded 86 complete, balanced within-student blocks, with 86 rating instances per configuration.

Regarding the distribution of question contexts, 280 instances (corresponding to 157 unique questions) used Mermaid flow-graph contexts (`Graph`), whereas 64 instances (corresponding to 33 unique questions across 13 repositories) used raw source-code contexts (`Raw`). Thirty-two students encountered at least one raw-code question, and 26 of these students answered questions under both context types, enabling within-student comparisons across context types.

4.4 Statistical Analysis Methods

To address our research questions while accounting for dependencies among observations from students, questions, and shared project repositories, we conducted statistical analyses using Python (`statsmodels`) [30]:

Scores and Composite Summaries. We calculated an exploratory multidomain composite, termed the *AI Agreement Index* (AAI), as the mean of all fifteen positively keyed items. Individual items, such as `ACCEPTANCE`, were analyzed as direct measures of specific perceptions.

Table 1. Survey instrument (translated from Vietnamese into English), comprising 15 positively keyed Likert items grouped into six conceptual constructs.

Construct	Item Code	Statement
Information & Pedagogical Quality	ACCURACY	The AI's comment correctly identifies the problems and errors in my answer.
	SPECIFICITY	The AI points exactly to which part of my answer needs revisiting, rather than commenting in general.
	CLARITY	The wording and presentation of the AI's comment are clear and unambiguous.
Explainability	TRANSPARENCY	I understand which academic criteria and evidence the AI used to assess my answer.
Usefulness & Actionability	USEFULNESS	This AI tool serves as an effective learning assistant for my project work.
	ACTIONABILITY	After reading the feedback, I know exactly which technical steps to take to improve my answer.
Trust, Reliability & Fairness	COMPETENCE TRUST	I trust the technical expertise and analytical capability of this AI system.
	RELIABILITY	The AI assesses consistently and does not change its grading standards at random.
	FAIRNESS	The AI's assessment fairly reflects the effort and actual quality of my answer.
Affect & Safety	AFFECT	I feel motivated and encouraged to keep learning after reading the AI's comments.
	PSYCHOLOGICAL SAFETY	Receiving AI feedback reduces the psychological pressure compared to human peer or instructor critique.
Acceptance & Behavioral Intention	ACCEPTANCE	I agree with the AI's judgment regarding the strengths and weaknesses of my answer.
	PERSUASIVENESS	The arguments and evidence provided by the AI are highly persuasive, leaving me convinced without wanting to argue back.
	WILLINGNESS	Based on this comment, I am ready to start refactoring or revising my answer right away.
	BEHAVIORAL INTENTION	I intend to continue using this LLM system to receive feedback for future course projects.

RQ1 (Perceptions and Drivers). To examine contemporaneous within-student associations with acceptance and behavioral intention, we fitted student fixed-effects linear regression models by demeaning predictors and outcomes within each student:

$$y_{ij} - \bar{y}_j = \sum_{k=1}^{m} \beta_k (z_{kij} - \bar{z}_{kj}) + \sum_{c=2}^{4} \gamma_c (D_{cij} - \bar{D}_{cj}) + e_{ij}, \tag{1}$$

where i indexes feedback instances, j indexes students, y denotes the acceptance or behavioral intention outcome, and D_c represents feedback configuration indicators, with *Sandwich* as the reference category. Predictors (perceived quality, utility, trust, affect, and safety) were standardized using full-sample standard deviations before within-student demeaning.

RQ2 (Feedback Configuration Comparisons). Primary comparisons focused on paired contrasts among the four feedback formats. To account for clustering within shared repositories, we reported repository-level bootstrap confidence intervals based on 5,000 resamples with replacement. We supplemented these analyses with exploratory non-parametric Friedman tests [10], Kendall's W, and Holm-adjusted Wilcoxon signed-rank tests [14]. To control the family-wise error rate across all 90 pairwise item-level contrasts

(6 format pairs × 15 survey items), we constructed simultaneous 95% sensitivity bands using a centered, bootstrap-standardized maximum statistic. For AAI scores, we assessed equivalence using the two one-sided tests (TOST) procedure with equivalence margins of ±0.25, ±0.20, and ±0.10 [21].

RQ3 (Context Type Association). Because context type was determined by the complexity-aware router ($\tau = 10$) rather than randomly assigned, it was associated with repository complexity. To examine these contextual associations, we fitted exploratory student fixed-effects models incorporating context-type and feedback-configuration indicators, with uncertainty estimated using 5,000 cluster-level bootstrap resamples. We further assessed sensitivity to repository clustering by iteratively excluding one repository at a time and recomputing the point estimates.

5 Results

5.1 Reliability and Data Quality

The fifteen positively keyed survey items showed high internal consistency ($\alpha = 0.936$, $\omega_t = 0.939$). Internal consistency estimates for the item groups were $\alpha = 0.91$ for information and pedagogical quality, 0.78 for usefulness and actionability, 0.80 for trust, reliability, and fairness, and 0.78 for acceptance and behavioral intention. Affect and psychological safety showed lower internal consistency ($\alpha = 0.56$), providing limited support for treating these two items as a single unidimensional construct. However, internal consistency alone does not establish the validity of a multidimensional factor structure.

To examine potential response-pattern anomalies, we identified 64 instances (18.6%) exhibiting straight-lining, in which students assigned identical ratings to all fifteen items. These included 11 students (12.8%) who exhibited straight-lining across all four feedback instances. After excluding these flagged responses, the pooled internal consistency estimate was $\alpha = 0.90$. However, excluding responses based on this pattern alone does not establish whether the remaining responses were attentive. Because all fifteen items were positively keyed, with reverse-worded items omitted to minimize linguistic confusion [32], these pooled response-pattern statistics describe properties of the response distributions rather than providing psychometric validation or supporting inference based on independent observations.

5.2 RQ1: Item Ratings and Associations with Acceptance

Table 2 summarizes item ratings across all 344 feedback instances. Mean ratings were above the neutral midpoint (3.0) for every item. The exploratory multidomain composite, the AI Agreement Index (AAI), had a mean of 4.07 (SD 0.64, 95% repository-bootstrap CI [3.90, 4.23]). The highest agreement rates were observed for perceived usefulness (84.0%), accuracy (83.4%), specificity (83.4%), and fairness (81.4%). Explicit disagreement (ratings ≤ 2) did not exceed 7.3% for any of the fifteen items.

Direct agreement with AI judgments (`ACCEPTANCE`) was reported in 79.9% of instances. Agreement with future-use intention (`BEHAVIORAL INTENTION`) was observed

Table 2. Item ratings across 344 feedback instances. CI: 95% repository-bootstrap confidence interval for the mean (28 repositories). Agree/Disagree: percentage of ratings ≥ 4 / ≤ 2. *The unweighted mean calculated directly from the 15 rounded item means is 4.06, which differs slightly from the multi-item composite mean (4.07) due to rounding effects across individual item averages.

Item Code	Mean (SD)	95% CI	Agree (%)	Disagree (%)
ACCURACY	4.38 (0.96)	[4.14, 4.61]	83.4	5.5
SPECIFICITY	4.36 (0.94)	[4.10, 4.60]	83.4	5.2
CLARITY	4.33 (0.88)	[4.09, 4.56]	81.7	3.5
TRANSPARENCY	4.29 (0.95)	[4.03, 4.54]	79.9	4.9
USEFULNESS	4.09 (0.77)	[3.93, 4.24]	84.0	2.9
ACTIONABILITY	4.13 (0.88)	[3.90, 4.34]	80.8	4.9
COMPETENCE TRUST	4.17 (0.98)	[3.94, 4.41]	76.5	4.9
RELIABILITY	3.88 (0.80)	[3.73, 4.01]	78.5	4.9
FAIRNESS	4.15 (0.79)	[3.98, 4.31]	81.4	2.3
AFFECT	3.53 (0.87)	[3.38, 3.71]	41.9	4.9
PSYCHOLOGICAL SAFETY	3.83 (0.85)	[3.65, 3.99]	76.2	7.3
ACCEPTANCE	4.09 (0.82)	[3.90, 4.27]	79.9	3.5
PERSUASIVENESS	3.87 (0.91)	[3.68, 4.05]	72.1	6.4
WILLINGNESS	3.97 (0.69)	[3.86, 4.08]	81.4	2.3
BEHAVIORAL INTENTION	3.80 (0.85)	[3.65, 3.94]	68.0	5.2

Table 3. Student fixed-effects regressions (Eq. 1). Cells report outcome rating points per full-sample predictor SD [95% repository-bootstrap CI], based on 5,000 bootstrap replicates. Configuration indicators are included.

Predictor Construct	Acceptance (ACCEPTANCE)	Willingness (WILLINGNESS)	Behav. Intention (BEHAVIORAL INTENTION)
Information & Pedagogical Quality	0.19 [−0.03, 0.41]	0.03 [−0.23, 0.32]	0.10 [−0.14, 0.36]
Usefulness & Actionability	0.36 [0.20, 0.51]	0.14 [−0.06, 0.25]	0.21 [0.06, 0.35]
Trust, Reliability & Fairness	0.24 [0.09, 0.42]	0.06 [−0.08, 0.22]	0.13 [−0.14, 0.40]
Affect (AFFECT)	−0.09 [−0.20, 0.03]	0.27 [0.03, 0.46]	−0.05 [−0.20, 0.10]
Psychological Safety	−0.05 [−0.12, 0.00]	−0.03 [−0.17, 0.07]	0.07 [−0.09, 0.25]

in 68.0% of instances, whereas learning motivation (AFFECT) received agreement ratings in only 41.9% of instances, with 53.2% receiving neutral ratings. These percentages represent feedback-instance-level proportions rather than independent student-level estimates. The findings suggest that favorable perceptions of feedback quality do not necessarily correspond to equally high reported motivation or intention to continue using the system.

Within-Student Conditional Associations. Table 3 presents student fixed-effects regression coefficients (Eq. 1) with repository-bootstrap confidence intervals. Perceived utility was positively associated with acceptance and future-use intention, with confidence intervals excluding zero. Trust was positively associated with acceptance, whereas the confidence intervals for perceived quality included zero after adjustment for feedback configuration. Motivation was positively associated with willingness to revise, while the confidence intervals for psychological safety included zero across all three outcomes.

5.3 RQ2: Differences Between Feedback Configurations

Table 4 presents the paired contrasts among the four feedback configurations. All six 95% repository-bootstrap confidence intervals for direct acceptance (ACCEPTANCE) included

Table 4. All six paired feedback-configuration contrasts. Mean differences [95% repository-bootstrap CI for Acceptance; 90% CI for AAI equivalence assessment], based on 5,000 bootstrap replicates. Positive values favor the first configuration.

Contrast	Acceptance: Difference [95% CI]	AAI: Difference [90% CI]
Sandwich – Rubric	0.047 [-0.121, 0.185]	0.114 [0.008, 0.212]
Sandwich – Explainable	-0.012 [-0.165, 0.122]	0.056 [-0.028, 0.128]
Sandwich – Researching	-0.023 [-0.198, 0.136]	0.029 [-0.066, 0.110]
Rubric – Explainable	-0.058 [-0.204, 0.084]	-0.058 [-0.119, -0.003]
Rubric – Researching	-0.070 [-0.274, 0.120]	-0.085 [-0.162, -0.014]
Explainable – Researching	-0.012 [-0.176, 0.153]	-0.027 [-0.076, 0.025]

zero, providing no evidence of statistically significant pairwise differences at the nominal 5% level. Mean scores on the secondary AAI composite ranged from 4.01 (*Rubric*) to 4.12 (*Sandwich*). An exploratory Friedman test yielded $\chi^2(3) = 23.04$, $p < .001$, and Kendall's $W = .089$, whereas an omnibus comparison using a mixed-effects model with student-specific random intercepts yielded $p = .084$.

At the item level, *Sandwich* and *Researching* received higher mean actionability ratings (4.33 and 4.28, respectively) than *Rubric* (3.97) and *Explainable* (3.95). The Sandwich–Explainable contrast in actionability yielded a mean difference of 0.37 points (unadjusted 95% repository-bootstrap CI $[0.16, 0.57]$), with a simultaneous 95% sensitivity band of $[0.02, 0.72]$ across all 90 pairwise item-level contrasts (6 format pairs $\times$ 15 survey items). A leave-one-repository-out sensitivity analysis, in which each of the 28 repositories was excluded in turn, yielded point estimates ranging from 0.34 to 0.41, with no reversals in direction.

Equivalence testing of the secondary AAI scores using the two one-sided tests (TOST) procedure showed that all six repository-bootstrap 90% confidence intervals fell within the ± 0.25 equivalence margin (e.g., the Sandwich–Rubric interval was $[0.008, 0.212]$). These results support equivalence of the average composite differences within this margin, although equivalence was not established at the individual-item level. Additionally, an intraclass correlation analysis indicated substantial student-level clustering (ICC = .77), suggesting that differences between students accounted for a large proportion of the observed rating variance.

5.4 RQ3: Raw-Code versus Graph-Based Questions

Mean acceptance ratings were 4.20 for raw-code questions and 4.06 for graph-based questions. After adjusting for student fixed effects and feedback-configuration indicators, the estimated coefficient for raw-code context was $b = 0.034$ (95% repository-bootstrap CI $[-0.180, 0.257]$). The secondary AAI analysis yielded $b = 0.072$ (95% CI $[-0.025, 0.187]$). Within-student variation in context type was observed among 26 students across 12 repositories. Because context type was determined by the complexity-aware router ($\tau = 10$) rather than randomly assigned, business-function complexity remained confounded with context representation. Therefore, these estimates cannot be interpreted as isolated causal effects of context type.

Table 5. RQ3 outcomes by context type. Descriptive means are reported, while b_{raw} adjusts for student fixed effects and feedback configuration. CIs are based on repository-level bootstrapping.

Outcome Metric	Graph Context (n = 280)	Raw Code Context (n = 64)	b_{raw} [95% CI]
Acceptance (`ACCEPTANCE`)	4.06	4.20	0.034 [−0.180, 0.257]
AAI (Secondary Composite)	4.06	4.14	0.072 [−0.025, 0.187]

6 Discussion

Acceptance, Motivation, and Future-Use Intention. Across the survey responses, agreement with statements concerning technical accuracy and perceived usefulness was considerably higher than agreement with statements concerning learning motivation (`AFFECT`) and future-use intention (`BEHAVIORAL INTENTION`). Student fixed-effects regressions indicated that perceived utility was positively associated with acceptance and future-use intention, while trust was positively associated with acceptance. These findings are consistent with established technology acceptance and trust in automation frameworks [4,33,22]. Notably, motivation was positively associated with willingness to revise after adjustment for psychological safety and other covariates, suggesting that different dimensions of learner affect may be associated with different outcomes. However, these self-reported perceptions do not establish actual code revision behavior or objective improvements in learning outcomes. Appropriate reliance on AI feedback therefore requires further evaluation against independent measures of system reliability [22].

Configuration Bundles and Pedagogical Design. Although mean AAI scores differed relatively little across the four formative feedback configurations, item-level results revealed more pronounced differences. For example, a mean actionability difference of approximately 0.37 points was observed between the *Sandwich* and *Explainable* configurations. Configurations incorporating explicit next steps (*Sandwich* and *Researching*) received higher actionability ratings. However, these configurations also differed in feedback structure, scoring criteria, response length, and model orchestration. Consequently, the observed differences cannot be attributed to any single pedagogical component. Future controlled experiments should independently manipulate these components to identify their individual contributions. Furthermore, the role-decoupled *Researching* configuration generated substantially longer explanations without clear evidence of an AAI advantage over the single-model *Sandwich* configuration. This finding highlights the need to assess computational efficiency and cost-effectiveness using actual API token usage and cost records. Finally, the high student-level intraclass correlation (ICC = .77) underscores the importance of accounting for individual differences in baseline ratings when evaluating educational feedback systems.

Response Patterns and Measurement Challenges. The identification of straight-lining in 18.6% of feedback-rating instances highlights potential measurement challenges when students evaluate multiple feedback instances consecutively using multi-item questionnaires. Reverse-worded items were omitted to reduce potential linguistic confusion [32]; however, the extent to which this design choice affected response quality was not directly assessed. Response fatigue also remains a potential concern when students evaluate

lengthy feedback across multiple criteria [20]. Future deployments should consider shorter, targeted questionnaires, embedded attention checks, and behavioral interaction logs to better distinguish attentive engagement from potentially biased response patterns.

6.1 Threats to Validity

Internal Validity. Because each feedback configuration was applied to a distinct student answer addressing a different scenario question, potential confounding influences from question difficulty, cognitive load, and order effects cannot be entirely ruled out. Although random permutation reduced systematic bias in configuration assignment, it did not eliminate variation in question characteristics or answer quality. Furthermore, for RQ3, context representation was determined by the complexity-aware router ($\tau = 10$), leaving business-function complexity confounded with context type. Database records documented configuration assignments but did not independently verify execution by external LLM providers.

Construct Validity. The survey instrument relied on self-reported perceptions without factor-analytic validation or independent assessment against expert evaluations of answer correctness. The secondary AAI composite assigned equal weight to conceptually distinct domains, while pooled internal consistency estimates combined within- and between-student variation. Additionally, concurrently collected self-reports may be subject to common-method variance [29]. Neither longitudinal learning gains nor actual code revision behavior was measured.

External Validity. The study was conducted at a single institution (FPT University), with voluntary participation in a capstone viva preparation setting. These characteristics may limit the generalizability of the findings to other institutions, mandatory assessment environments, or introductory programming courses. The findings may also depend on the characteristics of the participating repositories, scenario questions, and LLM configurations used in this study.

Statistical Conclusion Validity. Repository-level bootstrapping accounted for the clustering of observations within project repositories. However, with only 28 repository clusters, the resulting percentile-bootstrap confidence intervals remain approximate. The regression models captured contemporaneous within-student associations rather than causal relationships. Moreover, the ± 0.25 AAI equivalence margin, which was not independently validated against educationally meaningful differences, supports conclusions only within the chosen threshold rather than establishing definitive educational equivalence.

7 Conclusion and Future Work

Summarize the main takeaways, lessons learned, and future research directions.

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A Survey Instrument Details

Detailed survey questions and Likert-scale coding schemes can be provided here. Continues the same formatting standards as the main text.

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